Pizzeria Queries

Time limit: 1.00 s **Memory limit:** 512 MB

There are n buildings on a street, numbered $1, 2, \ldots, n$. Each building has a pizzeria and an apartment.

The pizza price in building k is p_k. If you order a pizza from building a to building b, its price (with delivery) is $p_a + |a - b|$.

Your task is to process two types of queries:

1. The pizza price p_k in building k becomes x.
2. You are in building k and want to order a pizza. What is the minimum price?

Input

The first input line has two integers n and q: the number of buildings and queries.

The second line has n integers $p_1, p_2, \ldots, p_n$: the initial pizza price in each building.

Finally, there are q lines that describe the queries. Each line is either "1 k x" or "2 k".

Output

Print the answer for each query of type 2.

Constraints

- $1 \le n, q \le 2 \cdot 10^5$
- $1 \le p_i, x \le 10^9$
- $1 \le k \le n$

Example

Input	Output
6 3 8 6 4 5 7 5 2 2 1 5 1 2 2	5 4